PRAVAN OMPRAKASH

St. Louis, MO | pravanop@gmail.com | pravanop.github.io/pravan-professional-website | LinkedIn | GitHub

SUMMARY

Research engineer and computational materials scientist who turns scientific questions into validated machine-learning models, reproducible Python software, and peer-reviewed results. Own projects from problem formulation and data generation through model development, uncertainty analysis, visualization, and publication. Lead developer of open-source tools for materials discovery, with expertise spanning DFT, statistical thermodynamics, and atomistic simulation.

TECHNICAL SKILLS

Programming & scientific computing: Python, NumPy, Pandas, SciPy, Matplotlib, Plotly, Jupyter, Pymatgen
Machine learning: PyTorch, TensorFlow, scikit-learn, graph neural networks, CNNs, transfer learning, uncertainty quantification
Computational science: Density functional theory (VASP), molecular dynamics (LAMMPS), statistical thermodynamics, phase stability, inverse design

EXPERIENCE

PhD Candidate & Graduate Researcher | Washington University in St. Louis — **Jan 2023-Present**
MCUBE Lab | St. Louis, MO

- Lead the end-to-end development of RAPTor, an open-source Python platform for phase diagrams, spinodal analysis, and alloy stability; benchmarked against 100+ experimental systems (81% accuracy) and used to generate predictions for 6,000+ alloys.
- Translate physical models into reusable computational workflows that combine DFT data, statistical thermodynamics, and machine learning to predict composition-temperature stability in high-entropy alloys.
- Co-designed SymPlex, a symmetry-conserving representation and visualization framework for high-dimensional alloy spaces; lead developer and co-first author in Scripta Materialia.
- Predicted hole doping lowers the coercive field of ferroelectric HfO_2 from 8 to 6 MV/cm and reduces a switching barrier from 180 to 80 meV/f.u.; first-author study in Physical Review Materials (2026).
- Work across theory and experiment to explain observed materials behavior, identify testable design rules, and communicate results through software, publications, and conference presentations.

Research Intern | Indian Institute of Science — **Jun 2021-Mar 2022**
Bengaluru, India

- Developed graph and convolutional neural networks for material potential-energy surfaces used in molecular-dynamics simulations, including atomistic-data preparation, model evaluation, and uncertainty estimation.

Research Intern | IIT Bhubaneswar — **Jan 2021-Feb 2022**
Bhubaneswar, India

- Built deep-learning surrogate models for inverse design of mechanical metamaterials and optimization of piezoelectric vibration-energy harvesters; translated finite-element simulation data into fast design-space exploration tools.

SELECTED MACHINE-LEARNING SYSTEMS

- Perovskite property prediction - curated a cross-chemistry crystal dataset and trained graph-representation models for band-gap prediction (0.28 eV MAE; millisecond inference); first author in Computational Materials Science.
- Chest X-ray opacity estimation - contributed to a deep transfer-learning pipeline combining lung segmentation, data balancing, model selection, cross-validation, and heatmap-based evaluation; published in Bioelectronic Medicine.
- AuthNet - co-developed a language-independent, contactless facial-authentication system using identity and temporal facial movements (98.1% test accuracy); AAAI 2021 Top 20 Student Abstract.

EDUCATION

PhD, Materials Science and Engineering | Washington University in St. Louis — **Expected Dec 2026**
St. Louis, MO | GPA: 3.75/4.00

BTech, Metallurgical and Materials Engineering | National Institute of Technology Karnataka — **2018-2022**
Surathkal, India | GPA: 8.6/10

SELECTED PUBLICATIONS & RECOGNITION

- Omprakash, P. et al. “Hole-doping reduces the coercive field in ferroelectric hafnia.” Physical Review Materials 10, 034416 (2026).
- Cavin, J.*, Omprakash, P.* et al. “SymPlex Plots for Visualizing Properties in High-Dimensional Alloy Spaces.” Scripta Materialia (2025). *Equal contribution.
- Swamynadhan, M. J. et al. “Design Rules and Discovery of Face-Sharing Hexagonal Perovskites.” Chemistry of Materials 38, 1679-1688 (2026).
- Best Oral Presentation, Materials Research Society Fall Meeting (2025), for work on interpretable visualization of multicomponent alloy spaces.